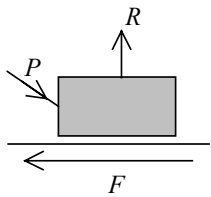


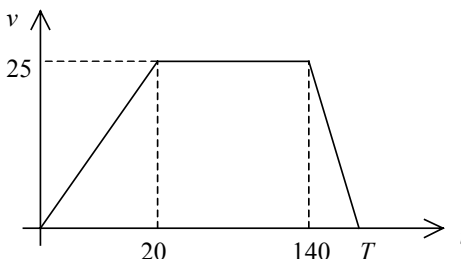
EDEXCEL MECHANICS M1 (6677)
PROVISIONAL MARK SCHEME NOVEMBER 2003

Question Number	Scheme	Marks
1.	(a) $0^2 = u^2 - 2 \times 9.8 \times 40$ $\Rightarrow \underline{u = 28 \text{ ms}^{-1}}$ (b) $-28 = 28 - 9.8 \times t$ $\Rightarrow \underline{t = 5.7 \text{ or } 5.71 \text{ s}}$	M1 A1 A1 (3) M1 A1 $\sqrt{\wedge}$ A1 (3) 6
2.	 (a) $28800 = 2000(12 - v)$ $v = -2.4 \text{ ms}^{-1}$ Speed = $\underline{2.4 \text{ ms}^{-1}}$ (b) due west / $\leftarrow$ / reversed direction (o.e.) (c) T : $28800 = m(6 + 3.6)$ $\Rightarrow m = \underline{3000 \text{ kg}}$ OR $2000 \times 12 - 6 \times m = -2000 \times 2.4 + m \times 3.6$ $\Rightarrow m = 3000 \text{ kg}$	M1 A1 A1 (3) A1 $\sqrt{\wedge}$ (1) M1 A1 M1 A1 (4) M1 A1 $\sqrt{\wedge}$ M1 A1 8

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3.	 $R \uparrow: R = 50g + P \sin 30^\circ$ $R \rightarrow: F = P \cos 30^\circ$ $F = \frac{3}{5}R$ used $P \cos 30^\circ = \frac{3}{5}(50g + P \sin 30^\circ)$ Elim F, R Solve <u>$P = 520$ or 519 N</u>	M1 A2, 1, 0 M1 A1 B1 M1 M1 A1 9
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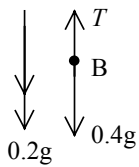
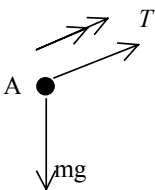
EDEXCEL MECHANICS M1 (6677)
PROVISIONAL MARK SCHEME NOVEMBER 2003

Question Number	Scheme	Marks
4.	(a)  (b) $\frac{1}{2}(T + 120) \times 25 = 4000$ $\left[\text{or } \frac{1}{2} \cdot 20 \cdot 25 + 120 \cdot 25 + \frac{1}{2}(T - 140) \cdot 25 = 4000 \right]$ $\rightarrow T = \underline{200 \text{ s}}$ (c) Car: $\frac{1}{2} \cdot 20 \cdot 25 + 25(t - 20) = 1500$ $\rightarrow t = 70 \text{ s}$ Hence motorcycle travels for 60s (d) $1500 = \left(\frac{0 + v}{2} \right) \cdot 60$ $v = \underline{50 \text{ ms}^{-1}}$	Shape B1 Figs B1 (2) M1 A1 A1 (3) M1 A1, A1 M1 A1 (5) M1 A1 (2) 12
5.	(a) $a = \frac{1}{4}[(5\mathbf{i} + 11\mathbf{j}) - (3\mathbf{i} - 5\mathbf{j})] = -2\mathbf{i} + 4\mathbf{j}$ (b) $\mathbf{F} = m\mathbf{a} = -6\mathbf{i} + 12\mathbf{j}$ $ \mathbf{F} = \sqrt{180} \approx 13.4 \text{ N (AWRT)}$ [OR $\mathbf{a} = \sqrt{20} \approx 4.47 \Rightarrow \mathbf{F} = 3 \times 4.47 \approx 13.4 \text{ N}$]	M1 A1 (2) M1 A1 M1 A1 (4)

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	(c) $t = 6 \quad \mathbf{v} = 3\mathbf{i} - 5\mathbf{j} + 6(-2\mathbf{i} + 4\mathbf{j}) \quad [= -9\mathbf{i} + 19\mathbf{j}]$ At B: $\mathbf{r} = (6\mathbf{i} - 29\mathbf{j}) + 3(-9\mathbf{i} + 19\mathbf{j}) \quad [= -21\mathbf{i} + 28\mathbf{j}]$ $OB = \sqrt{(21^2 + 28^2)} = \underline{35 \text{ m}}$	M1 A1✓ M1 A1✓ M1 A1✓ (6) 12
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EDEXCEL MECHANICS M1 (6677)
PROVISIONAL MARK SCHEME NOVEMBER 2003

Question Number	Scheme	Marks
6.	(a) M(D): $160 \times 2.5 = W \times 4 + 200(4 - x)$ $400 = 4W + 800 - 200x$ $200x - 4W = 400 \Rightarrow 50x - W = 100 *$ (b) M(D): $50 \times 2.5 + W \times 1 = 200(4 - x)$ $200x + W = 675$ (c) Solving $\rightarrow x = \underline{3.1\text{m}}$ $\therefore W = \underline{55\text{N}}$	M1 A2, 1, 0 M1 A1 (5) M1 A2, 1, 0 (3) M1 A1 M1 A1 (4) 12
7.	(a)  $0.4g - T = 0.4 \times \frac{1}{5}g$ (b) $T = \frac{8}{25}g$ or 3.14 or 3.1 N (c) $T - mg \sin 30^\circ = m \times \frac{1}{5}g$ $\rightarrow m = \frac{16}{35} *$  (d) Same T for A & B (e) $v^2 = 2 \times \frac{1}{5}g \times 1$ $v = \sqrt{\frac{2g}{5}} \simeq \underline{1.98 \text{ or } 2 \text{ ms}^{-1}}$	M1 A1 (2) M1 A1 (2) M1 A1 M1 A1 (4) B1 (1) M1 A1 (2)

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	(f) A: $-\frac{1}{2}mg = ma \Rightarrow a = -\frac{1}{2}g$ $v^2 = \frac{2g}{5} - 2 \times \frac{1}{2}g \times 0.4$ $\Rightarrow v = 0$	M1 A1 M1 A1 $\sqrt{\quad}$ A1 (5) 16
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